

1. Administrative & Study Identification

Full Study Title (Official Title): A Phase III, International, Multicenter, Randomised Open Label Study to Evaluate the Efficacy and Safety of Obinutuzumab Versus MMF in Patients With Childhood Onset Idiopathic Nephrotic Syndrome **Study Title:** A Study to Evaluate the Efficacy and Safety of Obinutuzumab Versus MMF in Participants With Childhood Onset Idiopathic Nephrotic Syndrome **Short Title/Acronym:** INShore **Protocol Number:** WA43380 (Other IDs: 2022-000369-42, 2023-505140-19-00) **NCT Number:** NCT05627557 **Posting ID:** 623296193197921049 **Sponsor/Company:** Roche (Hoffmann-La Roche) **Investigational Product:** Obinutuzumab (anti-CD20 monoclonal antibody) **Phase of Development:** Phase 3 **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of obinutuzumab compared with mycophenolate mofetil (MMF), as measured by the percentage of participants with sustained complete remission at 1 year without requiring systemic corticosteroids or rescue medications. **Secondary Objectives:** To assess overall relapse-free survival (RFS), probability of RFS at Week 52, and cumulative corticosteroid dose, alongside evaluating the safety, pharmacokinetics (PK), and pharmacodynamics (PD) of obinutuzumab compared with MMF.

2.2 Background & Rationale Childhood-onset idiopathic nephrotic syndrome causes the body to pass excess protein in the urine, leading to swelling and potential kidney damage. The current standard of care consists of a combination of steroids and immunosuppressive drugs like MMF. However, many patients suffer from frequently relapsing nephrotic syndrome (FRNS) or steroid-dependent nephrotic syndrome (SDNS) and experience disease recurrence. Additionally, prolonged steroid use carries the risk of severe side effects. Obinutuzumab targets CD20 proteins on B cells to provide a more targeted approach to reducing inflammation and protein loss in the urine, aiming to deliver better long-term outcomes and a more durable remission.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Mycophenolate Mofetil - MMF) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment (\$N\$): 85 participants **Target Population:** Children and young adults (aged \$ \ge 2\$ to 25 years) **Number of Sites:** International, Multicenter **Estimated Study Start:** Q4 2022 / Q1 2023 **Trial Status:** Active, Not Recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Diagnosis of frequently relapsing nephrotic syndrome (FRNS) or steroid-dependent nephrotic syndrome (SDNS) before the age of 18 years. 2. Must be in complete remission defined by the absence of edema, UPCR ≤ 0.2 g/g at screening, and have three consecutive daily urine dipstick readings of trace or negative for protein within the week prior to randomization. 3. Must have had at least one relapse in the 6 months prior to screening, after discontinuation of or while receiving oral corticosteroids and/or immunosuppressive therapy to prevent relapses. 4. Participants having received cyclophosphamide in the 6 months prior to randomization must have experienced at least 1 relapse subsequent to cyclophosphamide discontinuation. 5. Estimated glomerular filtration rate (eGFR) within normal range for age. 6. For females of childbearing potential: participants who agree to remain abstinent (refrain from heterosexual intercourse) or use highly effective contraception, during the treatment period and for 18 months after the final dose of obinutuzumab and for 6 weeks after the final dose of MMF. 7. For males: participants who agree to remain abstinent (refrain from heterosexual intercourse) or use contraceptive methods, and agree to refrain from donating sperm during the treatment period and for 90 days after the final dose of MMF.

4.2 Exclusion Criteria 1. Secondary nephrotic syndrome. 2. History of steroid-resistant nephrotic syndrome. 3. History of genetic defects known to directly cause nephrotic syndrome. 4. Treatment with other immunosuppressive medications to prevent relapse, other than MMF or oral corticosteroids within 2 months prior to randomization. 5. Pregnancy or breastfeeding or intending to become pregnant during the study or within 18 months after the final dose of obinutuzumab, or within 6 weeks after the final dose of MMF. 6. Females of childbearing potential, including those who have had a tubal ligation, must have a negative serum pregnancy test result within 28 days prior to initiation of study treatment and a negative urine pregnancy test at Day 1, prior to randomization. 7. History of organ or bone marrow transplant. 8. Participation in another therapeutic trial within 30 days of enrollment or 5 half-lives of the investigational drug. 9. Intolerance or contraindication to study therapies. 10. Participants demonstrating prior treatment failure to MMF as defined by two or more relapses in any 6-month period of time while receiving MMF for at least a 6-month duration. 11. Participants in the judgment of the investigator likely to

require systemic corticosteroids for reasons other than idiopathic nephrotic syndrome during the study. 12. Active infection of any kind or any major episode of infection requiring hospitalization or treatment with IV anti-infective medications within 4 weeks prior to screening, or completion of oral anti-infectives within 2 weeks prior to randomization. 13. History of or currently active primary or secondary immunodeficiency, including known history of human immunodeficiency virus (HIV) infection and other severe Immunodeficiency blood disorders. 14. History of progressive multifocal leukoencephalopathy. 15. History of or current cancer, including solid tumors, hematological malignancies, and carcinoma in situ within the past 5 years. 16. Major surgery requiring hospitalization during the 4 weeks prior to screening or during screening. 17. High risk for clinically significant bleeding or any condition requiring plasmapheresis, intravenous immunoglobulin, or acute blood product transfusions. 18. Evidence of any significant or uncontrolled concomitant disease that, in the investigator's judgment, would preclude participant's participation, including but not limited to nervous system, respiratory, cardiac, hepatic, endocrine, malignant, or gastrointestinal disorders. 19. Currently active alcohol or drug abuse or history of alcohol or drug abuse.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Experimental Arm:** Obinutuzumab 1000 mg (or 20 mg/kg for participants ≤ 45 kg) administered on Days 1, 15, 168 (Week 24), and 182 (Week 26). Premedication includes Methylprednisolone, Diphenhydramine HCl (0.5-1 mg/kg), and Paracetamol/Acetaminophen (15 mg/kg). * **Active Comparator Arm:** Mycophenolate Mofetil (MMF). **Route of Administration:** Intravenous (IV) infusion for Obinutuzumab; Oral (PO) for MMF and premedications. **Duration of Treatment:** 52 weeks (1 year).

5.2 Concomitant & Prohibited Therapy Permitted: Prednisone or equivalent at randomization following a guided tapering schedule to reach a goal of 0 mg/day by Weeks 4-6 (no later than Week 8), with the aim to continue without prednisone through Week 52. **Prohibited:** Other immunosuppressive medications to prevent relapse (other than MMF or oral corticosteroids) within 2 months prior to randomization; live virus vaccines; other investigational therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day -1 Informed Consent, Medical History, Urine Dipstick, UPCR, Infection Screening
Baseline	Day 1	Randomization, First Dose of Obinutuzumab	

(Group A) or MMF (Group B), PK/PD sampling | | **Interim** | Day 15, Week 24, Week 26 | Obinutuzumab subsequent IV infusions (Days 15, 168, 182), Efficacy/Safety Labs | | **End of Study** | Week 52 (1 Year) | Primary Endpoint Assessment (Complete Remission Check), Safety Labs |

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. **Percentage of Participants with Sustained Complete Remission at 1 year.** Complete remission is defined as undetectable protein in the urine, with no relapses during the treatment period that require the use of systemic corticosteroids or rescue medications.

7.2 Secondary Outcomes 1. **Overall Relapse-free Survival (RFS)** 2. **Probability of RFS at Week 52** 3. **Cumulative Corticosteroid Dose** (*Exploratory safety and PK/PD parameters for obinutuzumab will also be gathered*).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via site visits and patient reporting, particularly monitoring for infusion-related reactions (mitigated by premedication), neutropenia, and hypogammaglobulinemia. **Serious Adverse Events (SAEs):** Monitoring for severe infections, primary or secondary immunodeficiency complications, and progressive multifocal leukoencephalopathy (PML). **Data Monitoring Committee (DMC):** Internal and/or External safety monitoring mechanisms managed by Roche.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all patients receiving at least one dose of the study drug. **Sample Size Justification:** Powered to detect a statistically significant difference in complete remission rates at 1 year between the obinutuzumab and MMF arms (Target Enrollment: 85). **Handling of Missing Data:** Standard multiple imputation or model-based methods suitable for open-label, randomized efficacy trials. Dropouts due to treatment failure or requiring rescue medications are considered as not achieving complete remission.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and centralized monitoring by the sponsor to verify adherence to the protocol and accurate recording of efficacy/safety endpoints. **Audit Trail:** Secure eCRF systems with audit trails for all entered data, including lab values and clinical assessments.

11. Study Identifiers & Governance

Primary ID: WA43380 **Registry Number:** NCT05627557 **Posting ID:** 623296193197921049 **Sponsor/Lead Organization:** Roche **Company:** Roche **Study Title:** A Study to Evaluate the Efficacy and Safety of Obinutuzumab Versus MMF in Participants With Childhood Onset Idiopathic Nephrotic Syndrome (INShore Study) **Official Regulatory Title:** A Phase III, International, Multicenter, Randomised Open Label Study to Evaluate the Efficacy and Safety of Obinutuzumab Versus MMF in Patients With Childhood Onset Idiopathic Nephrotic Syndrome

12. Clinical Framework (INShore Specific)

Disease Area / Condition: Childhood Idiopathic Nephrotic Syndrome (FRNS/SDNS) **Diagnosis Threshold:** History of disease onset before age 18; relapse history requiring steroids/immunosuppressants; UPCR ≤ 0.2 g/g at baseline for remission. **Medical Methodology:** Anti-CD20 directed B-cell depletion vs. Standard Immunosuppression **Optimization Target:** Maintenance of complete remission without the need for systemic corticosteroids.

13. Study Procedures & Methodology

Management Pathway: Targeted monoclonal antibody therapy vs. oral active comparator. **Education Protocol (HCP):** Site initiation and training on IV infusion protocols and premedication requirements. **Education Protocol (Patient):** Counseling on the importance of strict tapering of prednisone to 0 mg by weeks 4-8. **Quality Audit Mechanism:** Real-time monitoring of dipstick diaries and UPCR laboratory results.

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks of primary treatment and observation **Onsite Visit Cadence:** Frequent visits during the infusion weeks (Days 1, 15, Week 24, Week 26) and tapering period. **Remote Monitoring Frequency:** As needed based on clinical status and local site protocols. **Checklist Review Interval:** Regular review of home urine dipstick testing for relapse detection.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]					